MATS 12 · NEEL NANDA STREAM · EXPLORATION PHASE · 30 OCTOBER 2026

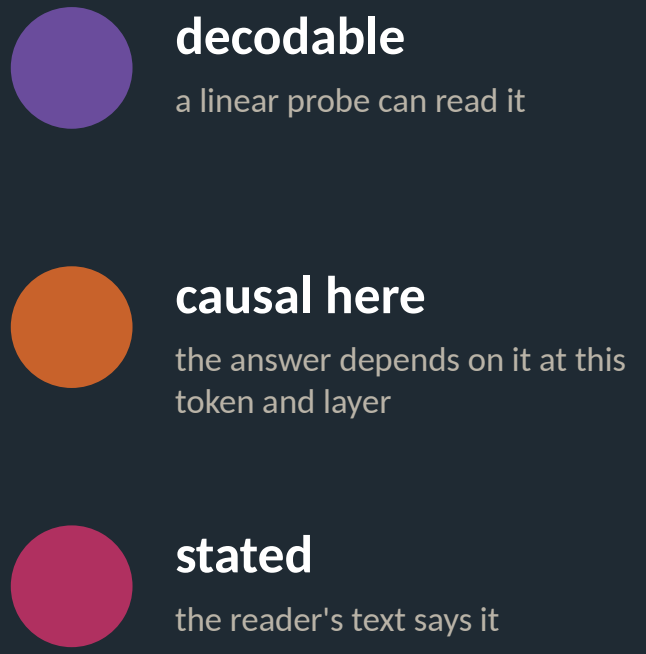
# When do meta-models reveal more than the prompt?

## Reporting gaps and read-site limits in Oracle Lens and NLA

*Working hypothesis: two activation readers mostly tell you what the prompt says, and report Qwen's own state only where it has become large at the read token.*

Qwen3.6-27B · synthetic banks and one public benchmark.

**Guiv Farmanfarmaian** · **Suraj Prasad**




*Our 3 main measurements*

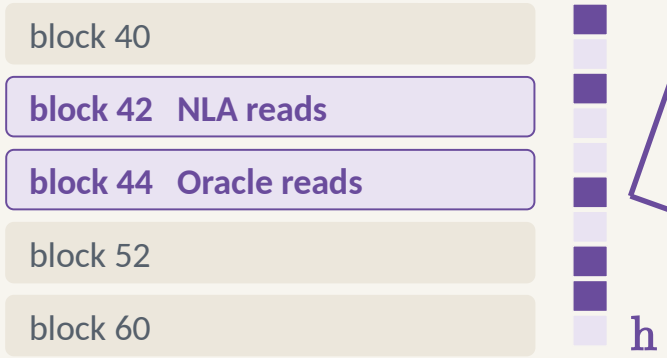
decodable · causal here · stated

# An activation reader turns one residual vector into text

### 1 Prompt into Qwen3.6-27B

```
Archived question: The capital
of Peru is
End of archived question.
Current question: The capital
of France is
```

### 2 Residual output of one block, one token




`block L output = hidden_states[L+1]`

### 3 Inject h into a reader that shares Qwen's weights

**Oracle Lens · reads block 44**

replaces a marker token's input embedding

$$e[\text{marker}] \leftarrow 16000 \cdot h \,/\, \|h\|$$

**NLA · reads block 42**

adds after its block 1, RL-trained so a critic can rebuild h

$$h_1[\text{marker}] \leftarrow h_1 + \|h_1\| \cdot h \,/\, \|h\|$$

### 4 Text

→ **short continuations of a prompt**

→ **a paragraph explaining the activation**

**Read site** Chat banks read the assistant-onset token, the last prompt token before Qwen would generate. Transfer chains (E133, E137) have no chat template and read the raw final " by". Never the first generated token.

Readers: Oracle Lens and NLA adapters on the same Qwen3.6-27B backbone · core/readers.py and read-site conventions in guivff/activation-readers @ c60fa970 (appendix A3)

decodable · causal here · stated

# Readouts often look like a prompt of this kind, rebuilt and answered

| Prompt to Qwen | Oracle Lens readout | NLA readout |
|---|---|---|
| What is the opposite of 'hot'? (Answer in Spanish, one word.) | - **Frío**<br>- Frío<br>- Frío (This is the single most common Spanish word for "cold" and the most direct translation) | Q&A format pattern: structured question-and-answer pair in English, now delivering a **Spanish-language direct answer**. … requiring the single Spanish word (e.g., "Frío") as the completion. |

lang-02A

*Both state the answer, and the NLA names the answer language.*

| | | |
|---|---|---|
| Elif carries the rope and Jonas carries the lantern. Who carries the lantern? Name only. | - **Alice**<br>-<br>**John**<br><br>*(Note: The answer is strictly the name of the person who sleeps, as requested by … | Q&A format with strict answer-only instruction signals the model must provide just the name **"Alex"** as the direct answer to "Who sits first." … |

own-02A

*Neither names Jonas or Elif. Both rebuild a prompt of this kind (a name-only question) and answer that.*

**Selected examples** Two readouts picked to show the two styles, not rates; the rates are on the following slides. Excerpts shortened with …; highlights mark the words a judge would credit.

G1 pilot, results/g_agreement/agam/viewer.html (items lang-02A, own-02A) · both readers at Qwen block 42 (off the Oracle's trained grid), last prompt token, first sample · guivff/activation-readers @ c60fa970

decodable causal here stated

# Three questions about every state: decodable, causal here, stated

## Decodable

**Can a linear probe read it from the activation?**

held-out probe

AUROC

0.5 chance · 1.0 perfect

*judge-free*

## Causal here

**Does Qwen's answer depend on it at this token and layer?**

patch the activation, watch the answer's logit difference

R

0 no effect · 1 full effect

*judge-free*

## Stated

**Does the reader's text say it?**

blind judge; a YES must quote the readout

k / n

missing readouts reported

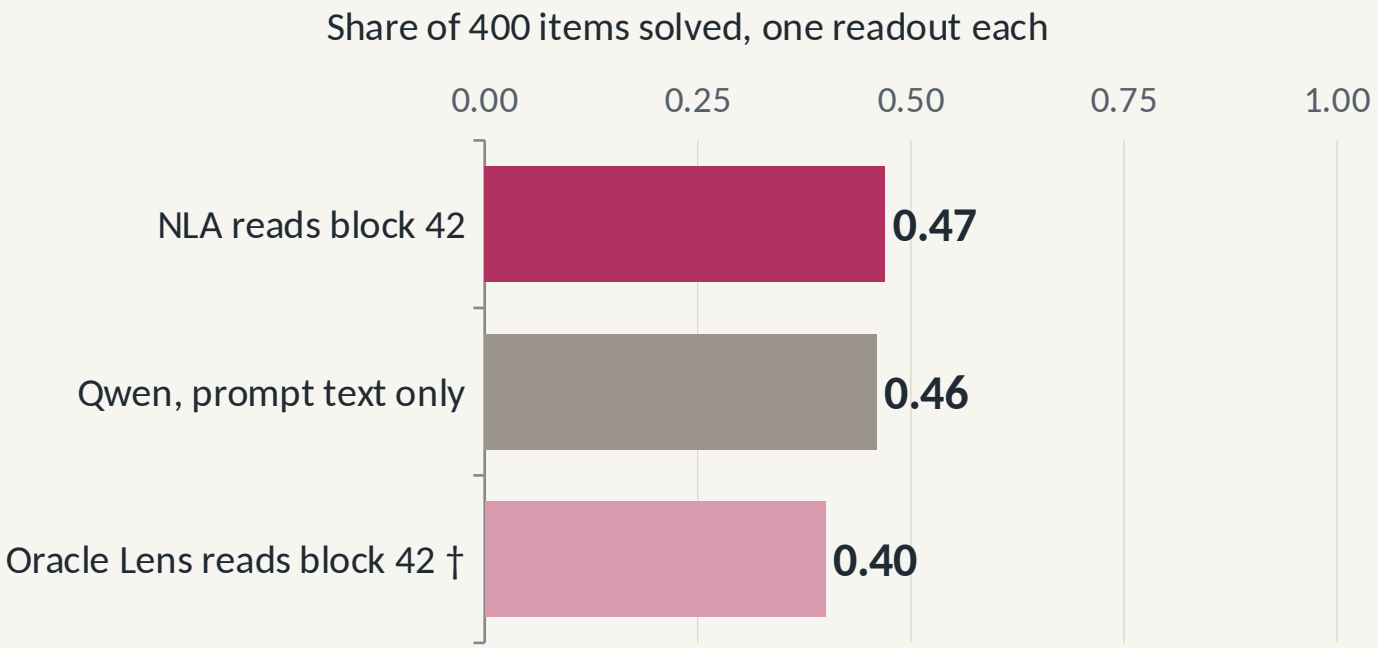
*judged*

**Rule** A state can be decodable without being causal here, and causal without being stated. These are never drawn on a shared axis; each later slide names its unit.

Judge: blind LLM judge, a YES must quote the readout · human audit E71 (κ 0.68 on 63 usable readouts, registered bar 0.70: NOT MET), appendix A14

decodable causal here stated

# On WorkspaceBench, the NLA scores what a text-only copy of Qwen scores




**95% CIs** NLA 0.42–0.52 · text-only 0.41–0.51 · Oracle 0.35–0.45 † off the Oracle's trained grid (0.37 / 0.39 at 40 / 44)

**NLA vs text-only: McNemar p = 0.75 (80 vs 75 discordant items)**

NOT COMPUTE-MATCHED

**Best-of-N scores, for reference**

**0.65** Oracle, published protocol: any of 11 layers (20–60) passes

**0.69** text-only Qwen, best of 3 samples

**0.96** any of 8 text-only samples (2 models × 2 templates)

**0 of 187** NLA hits that every text-only sample missed

**Does not show** That the reader ignores the activation (at this layer the activation encodes the prompt), or equivalence: p = 0.75 is a failure to find a difference, not a proof of none.

Suraj exp 06 (author result), 400 items, four multi-token families, item-bootstrap 95% CIs, McNemar on 80 vs 75 discordant items · regex hit instrument mt-regex-2026-09-23 · Jarus77/mats-nla-workspace @ 4d43ecb2

decodable · causal here · stated

# When Qwen's own answer changes with the wording, the reader rarely follows it

## The test · template shown, realised pairs are private

```
What is the capital city of {entity}?
```

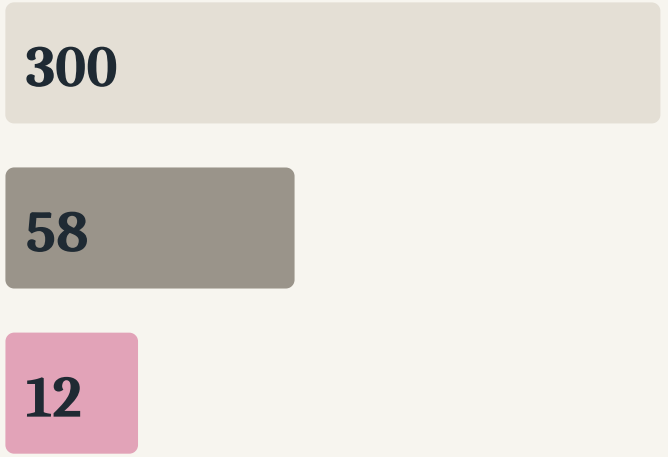
→ **Qwen answers X**

```
Which city is {entity}'s capital?
```

→ **Qwen answers Y (≠ X, by selection)**

Read both at block 60. Tracking: reader says X on wording A and Y on wording B. Same answer: one value for both.

## Eligibility funnel · reader at block 60




**Of 12: 1 follows Qwen, 0.08 [0.01, 0.35] · 11 give one answer for both, 0.92 [0.65, 0.99]**

**Does not show** That the reader re-solves in general: rates are conditional on 12 of 58 pairs (7/116 readouts missing, 15 state several values). Paraphrases also move text-only models; a small natural difference may be below what the reader registers.

E140 (descriptive pre-stated readings: tracking NOT MET, same-answer MET) · guivff/activation-readers @ c60fa970 · e140/analysis/REPORT.md, RESULTS.json · templates projects/g_agreement/e140/TEMPLATES.json · reviewed R-5

decodable · causal here · stated

# After a whole-activation swap, content transfers readily; structural reporting is sparse

## The intervention · Oracle Lens, block 44

recipient prompt

donor prompt

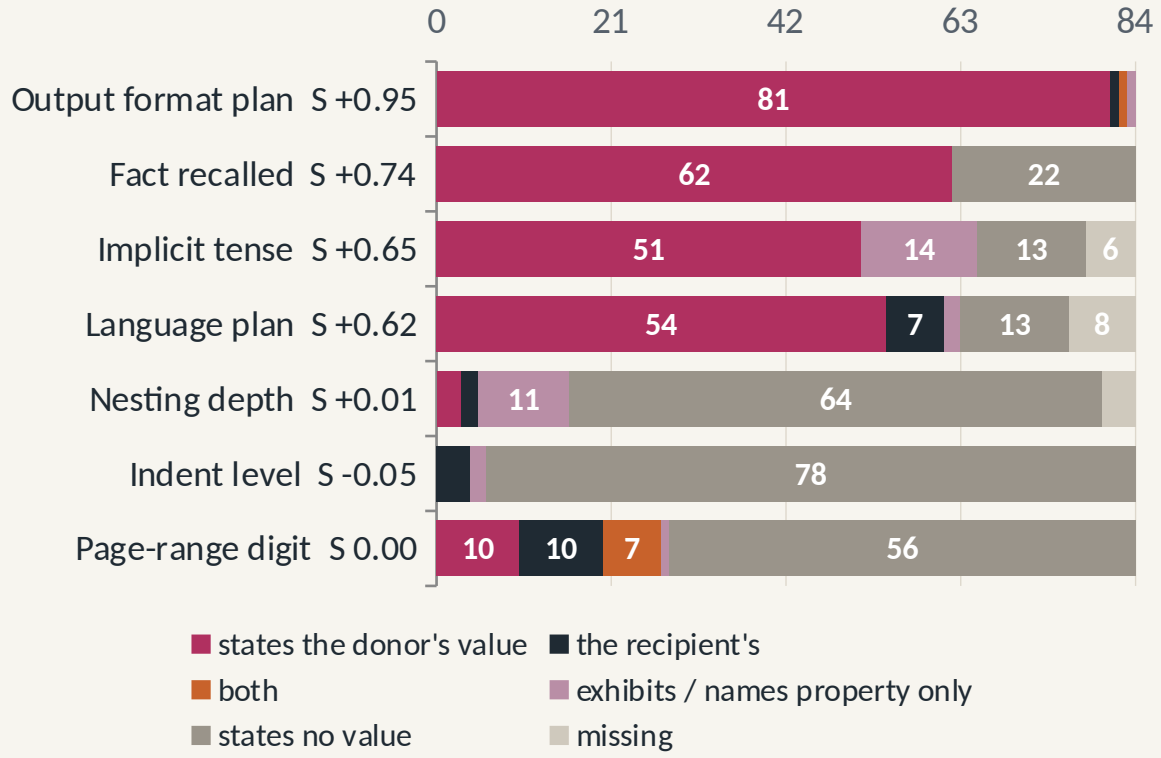
$h_{44}(\text{recipient}) \leftarrow h_{44}(\text{donor})$

→ reader writes text → judge: whose value does it state?

$$S = (\#\text{donor-only} - \#\text{recipient-only}) \,/\, \#\text{kept}$$

Content rows follow the donor. On nesting and indent, S ≈ 0 because the readout states no value at all (64 and 78 of 84). On the page digit it is partly cancellation: 10 donor, 10 recipient, 7 both.

## Oracle readouts after the swap · counts of 84 per row

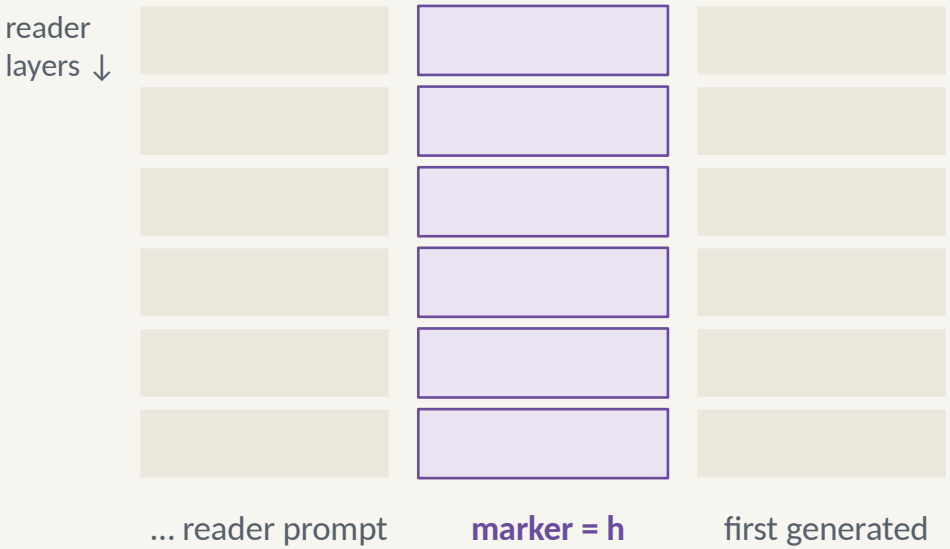



**Does not show** That the reader cannot see structure: probes read these rows at AUROC 1.00 (next slide). A transplant moves everything at once; the category census is post hoc.

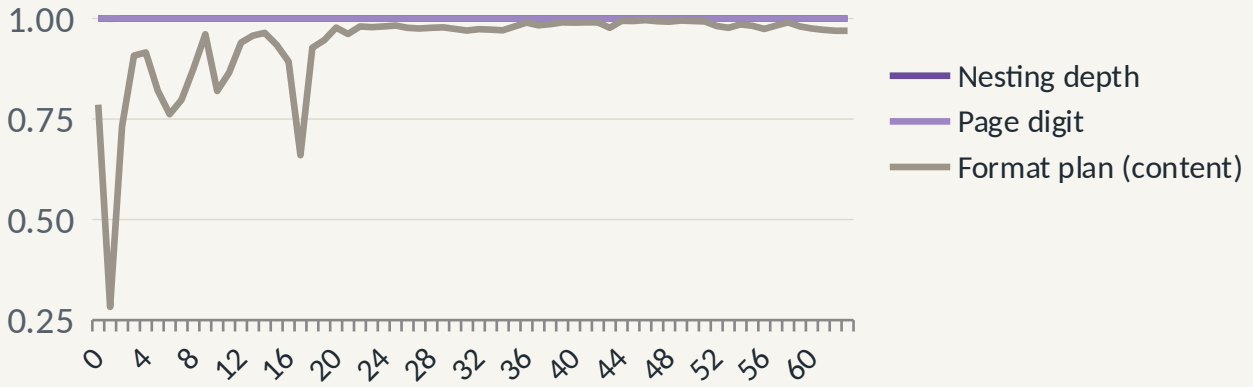
decodable · causal here · stated

# Inside the reader the structural state is perfectly decodable, but seldom stated

## Where the probe reads

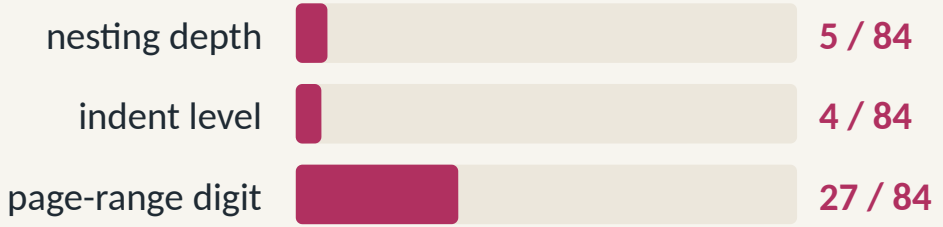



Decodable · TEST AUROC at the first generated token, reader block 0–63




Probe every reader block (0–63) at two positions: the injected marker and the first generated token. 84 dev / 84 test items per row; dev-only model selection. The first generated token is always the same token (ID 12), so token identity cannot explain it.

**Stated · in E100's readouts, the Oracle states any value of**




**Does not show** That the reader uses the state: at the injected marker the vector is its own input, so 1.00 there is trivial; the first-token probe shows the state reaches generation positions, not that it drives words.

decodable · causal here · stated

# A prompt-stated cue reaches the reader's word choice and is never named

## Follow one cue from Qwen's activation to the reader's words

| | | |
|---|---|---|
| **In Qwen's activation, block 42 and 44** | probe, 400 test | **AUROC 1.00** |
| ↓ | | |
| **Inside the reader, all 64 blocks** | probe, 2 positions | **128 / 128 at 1.00** |
| ↓ | | |
| **In the readout's word choice** | bag of words, grouped CV | **AUROC 0.988** |
| ↓ | | |
| **Stated by the Oracle reader** | blind judge, kept | **0 / 176 · 0 / 350** |

### Words the classifier leans on

**cue present**

department · recommends · last · incident · sentence

**cue absent**

mentioned · local · stated · park · visit

*The cue changes the register of the rebuild. Nothing says "evaluation".*

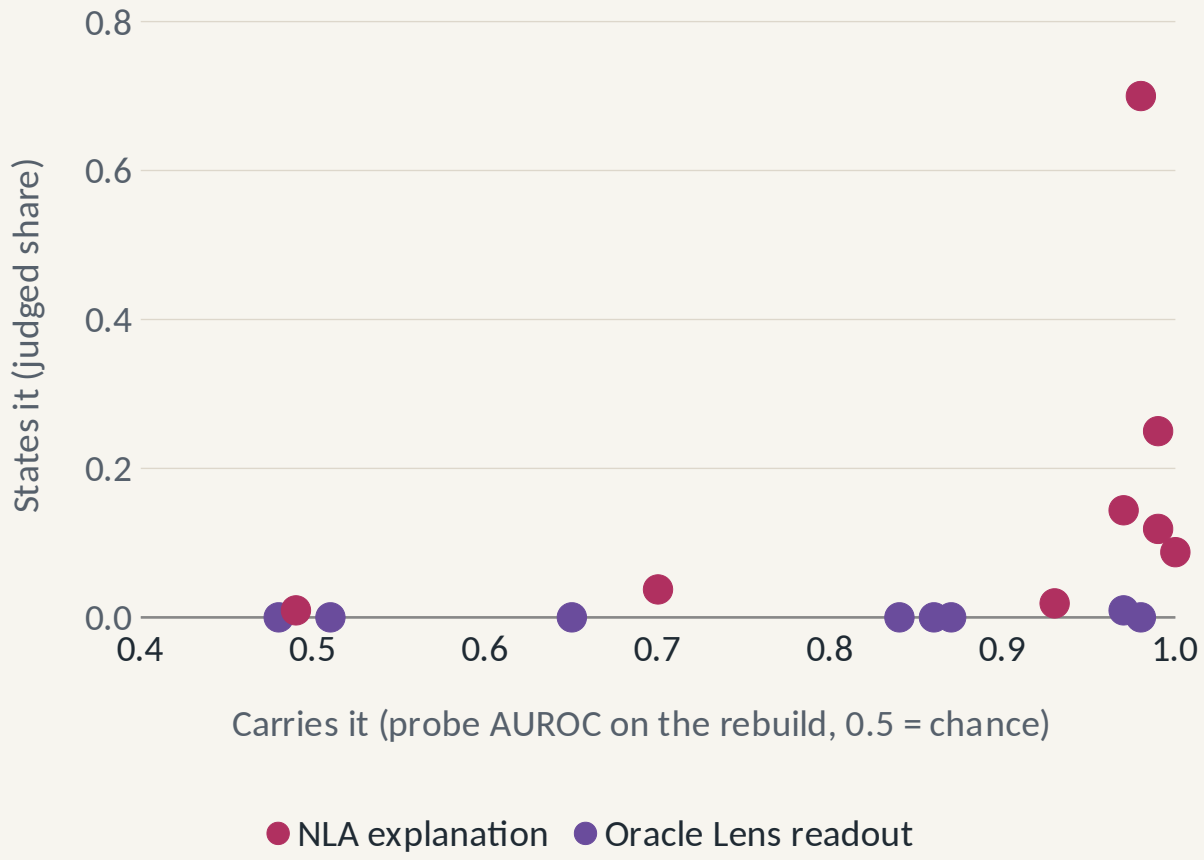
**Does not show** Evaluation awareness: the cue is written in the system prompt. A genre contrast is just as recoverable from the words (0.987), and asking the reader directly was not evaluable.

E85 (R-E1 NOT MET: every mention rate 0), E120a/b/c, E85 readout-word CV (exploratory) · reviewed R-2 · guivff/activation-readers @ c60fa970

PUBLIC CODE FILES, BOTH READERS

decodable causal here stated

# Readers' text often carries structural state that is rarely stated




| Row | Carries · Oracle / NLA | States · Oracle / NLA |
|---|---|---|
| **Python string** | 0.86 / 0.99 | 0.00 / 0.12 |
| **JS string** | 0.84 / 0.97 | 0.00 / 0.14 |
| **LaTeX math** | 0.87 / 1.00 | 0.00 / 0.09 |
| **Call argument** | 0.51 / 0.93 | 0.00 / 0.02 |
| **JSON depth** | 0.97 / 0.99 | 0.01 / 0.25 |
| **MD heading** | 0.65 / 0.70 | 0.00 / 0.04 |
| **MD list** | 0.48 / 0.49 | 0.00 / 0.01 |
| **Code language (control)** | 0.98 / 0.98 | 0.00 / 0.70 |

*Carried, not stated: the readout reproduces the text that holds the state, and does not name it.*

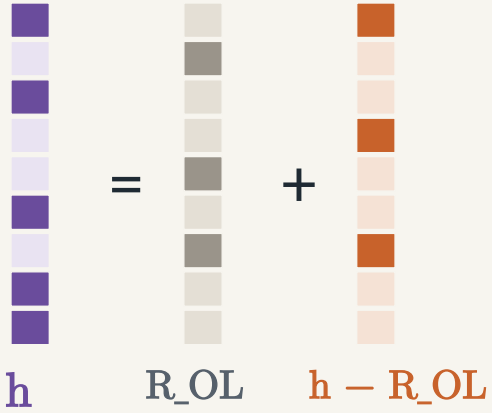
**Does not show** That the readers use the state, or that the Oracle carries less: its readouts are rebuilt by the NLA's critic, which is out of distribution for them. Carry and stated rates come from different splits of the same bank.

Suraj exp 25 (carries: results/table25.md, 160 items per row, k = 1) and exp 22 (states: results/score22.json, 40 items × 8 samples) · author results · Jarus77/mats-nla-workspace @ 4d43ecb2

decodable · causal here · stated

# Experiment S16: position-in-text state follows the part of the activation the Oracle's text does not rebuild

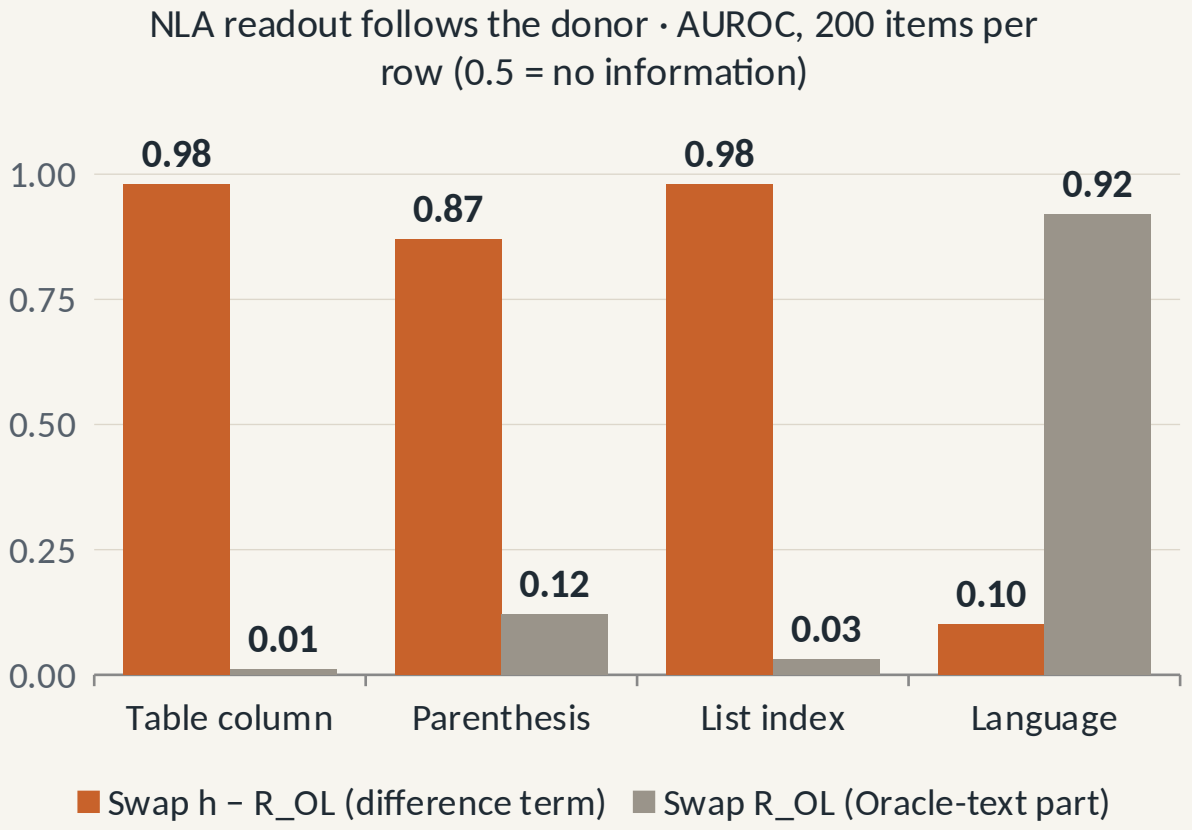
## Decompose the NLA-space activation at block 42




**R_OL** what the NLA's reconstructor rebuilds from the Oracle's text
**h − R_OL** the part that text fails to rebuild

Swap one part from a donor item, read with the NLA, ask whether the readout follows the donor

**Matched check (E7, other banks):** 0.39 · 0.57 · 0.51 on the same three rows. Unreconciled (E27).

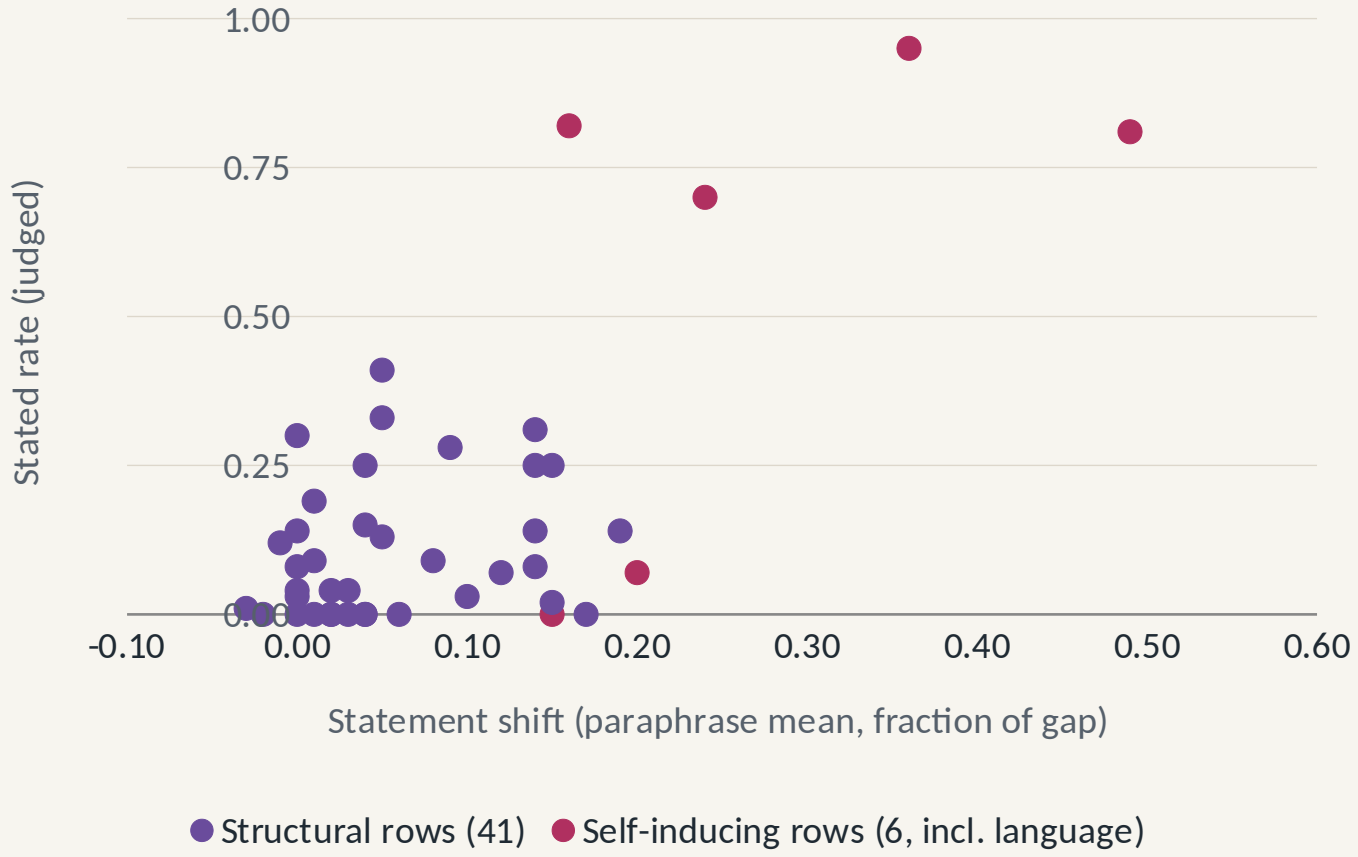



**Does not show** A mechanism. Guiv's matched swap (E7) on different banks found much weaker following (0.39 / 0.57 / 0.51); a dated reconciliation (E27) could not assign the gap to one cause.

Suraj exp 14 and 16 (author; S16 run by Guiv's agent on Suraj's pipeline), 200 recipient items per row · Jarus77/mats-nla-workspace @ 4d43ecb2 · E27 reconciliation, guivff/activation-readers @ c60fa970

decodable causal here stated

# How far a statement shifts the critic's reconstruction does not predict whether the NLA states structure




## The registered rule

*"Shift ≥ 0.30 ⇒ the NLA states it"*

**WITHDRAWN on 47 rows**

**Counter-rows** 7 stated ≥ 0.25 with shift < 0.17

**All rows** Spearman 0.40, p 0.006

**Structural only** 0.28, p 0.08

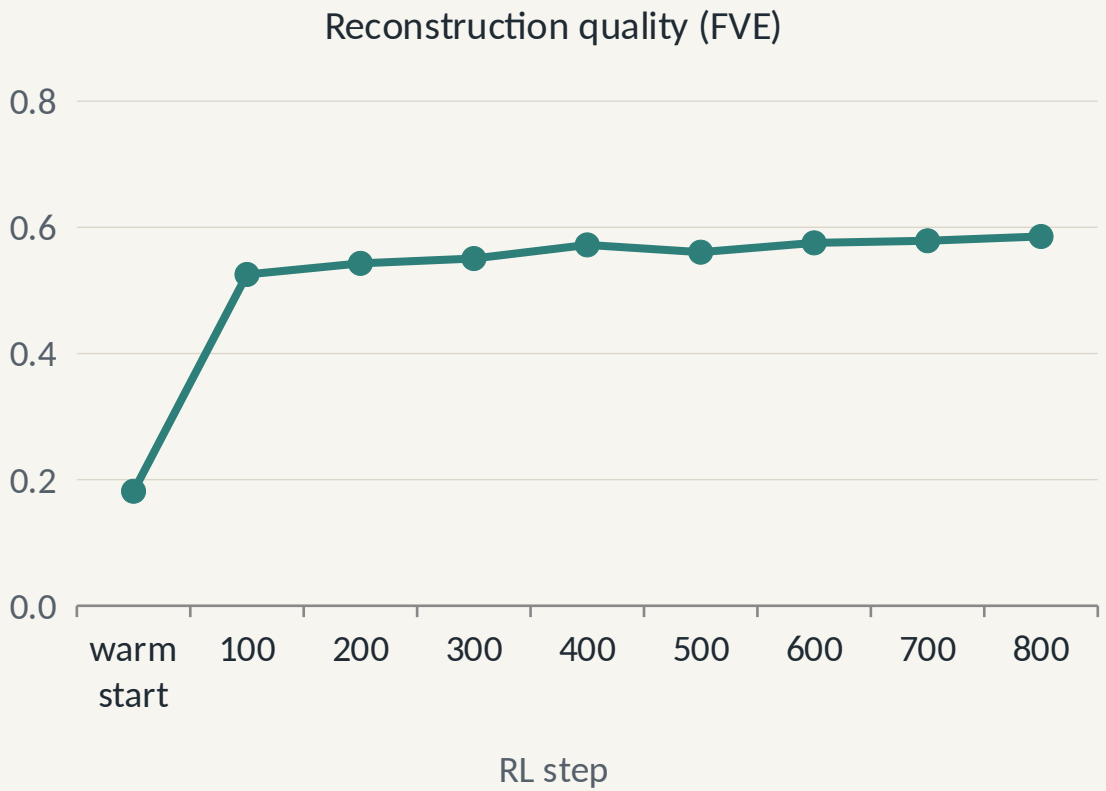
*Silence on structure looks like writing style, not a graded reward signal.*

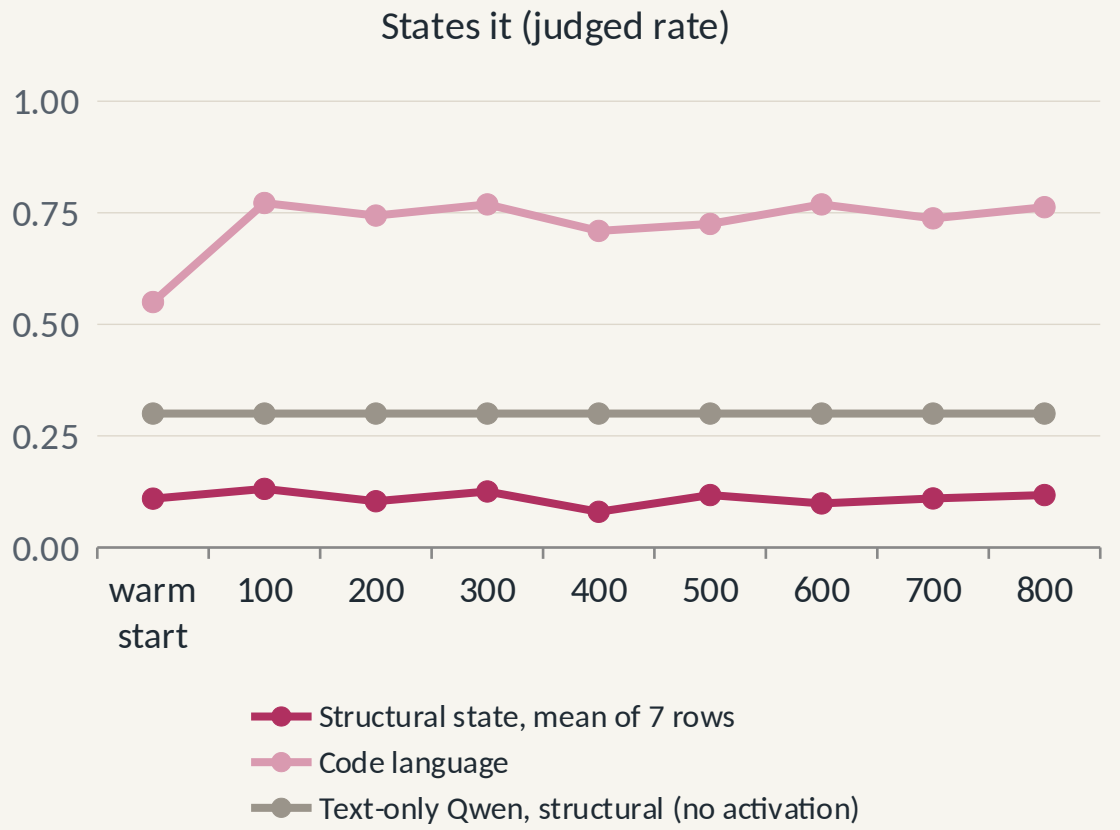
**Does not show** That the reconstruction objective is irrelevant, or anything about the full training reward: the shift is a geometric proxy. Inside structural rows the relation is weak (Spearman 0.28, p = 0.08, 41 rows); the steep part is driven by language-like rows.

Suraj exp 28 (author), 47 rows · x: paraphrase-mean statement shift, 8 paraphrase pairs · y: judged NLA stated rate over 320 readouts per row · Jarus77/mats-nla-workspace @ 4d43ecb2 · results/table28.md

decodable causal here stated

# The silence is there before RL; 800 RL steps raise reconstruction, but not structural statements



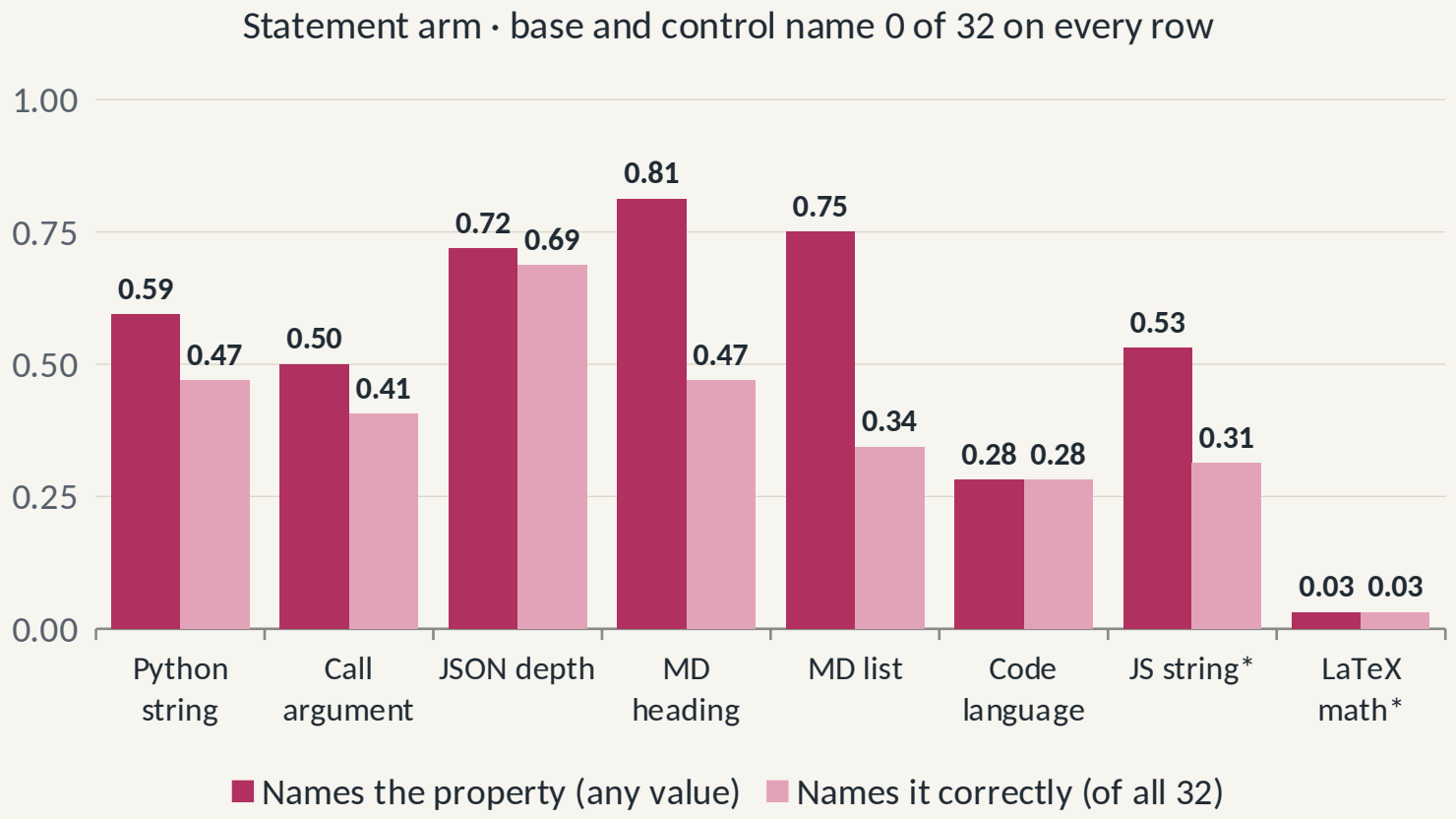




**Does not show** Whether the activation input or the warm-start style causes the drop from text-only: the text-only arm sees the prompt, not the activation, so the two are confounded.

Suraj exp 29 (author) · results/table29.json · judged rate over 320 readouts per row and checkpoint · FVE scored with the final critic for every checkpoint · Jarus77/mats-nla-workspace @ 4d43ecb2

decodable causal here stated

# Supervising statements makes the reader name structure; reconstruction barely moves, transfer is weak







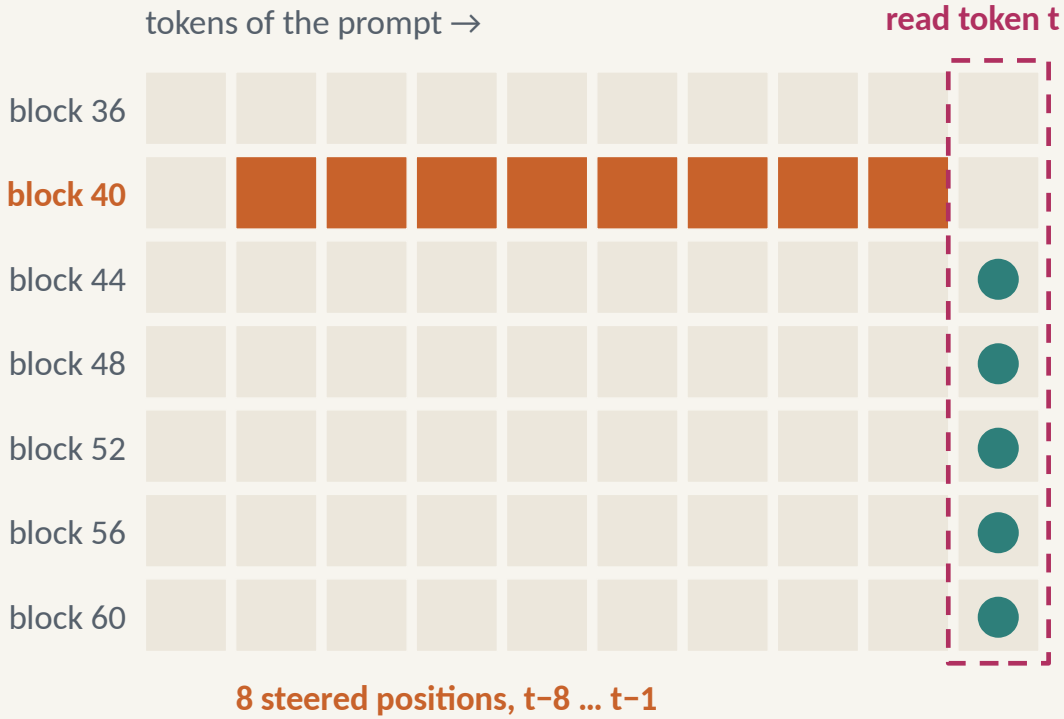
**P1 partly met**
within 0.02 of control: yes (0.005)
within 0.03 of 0.761: no (~0.25 below);
reference from exp 00, unreconciled

**Does not show** Broad repair, or that a reward change would do the same: this is supervision on six properties. P1's absolute FVE reference (0.761) is unreconciled; P2–P4 NOT MET, P5 not scorable.

Suraj exp 30 (author, pre-registered P1–P5) · eval30_{base,C,S}.json, 32 held-out items per row, k = 1 · gate FVE n = 192 per arm · * not trained · Jarus77/mats-nla-workspace @ 4d43ecb2

decodable · causal here · stated

# Push the answer upstream at block 40, then put one clean token back at a later block




steer: block 40, the 8 positions before t; t itself untouched

restore: clean activation into t at one block $\ell$, one run per $\ell$

## 1 Steer

$$\tilde{h}_{40,i} = h_{40,i} + \alpha \cdot \|d\| \cdot \hat{d} \qquad i = t-8 \ldots t-1$$

d: class-mean difference of block-40 activations on 84 dev items · at α = 8 the push is 2.23× the median block-40 norm

## 2 Score Qwen's answer

$$\Delta = \log p(\text{target continuation}) - \log p(\text{original})$$

fixed continuations: the steering target vs Qwen's original answer

## 3 Restore one cell, measure what is undone

$$R_\ell = (\Delta_\text{steered} - \Delta_\text{restored},\ell) \;/\; (\Delta_\text{steered} - \Delta_\text{clean})$$

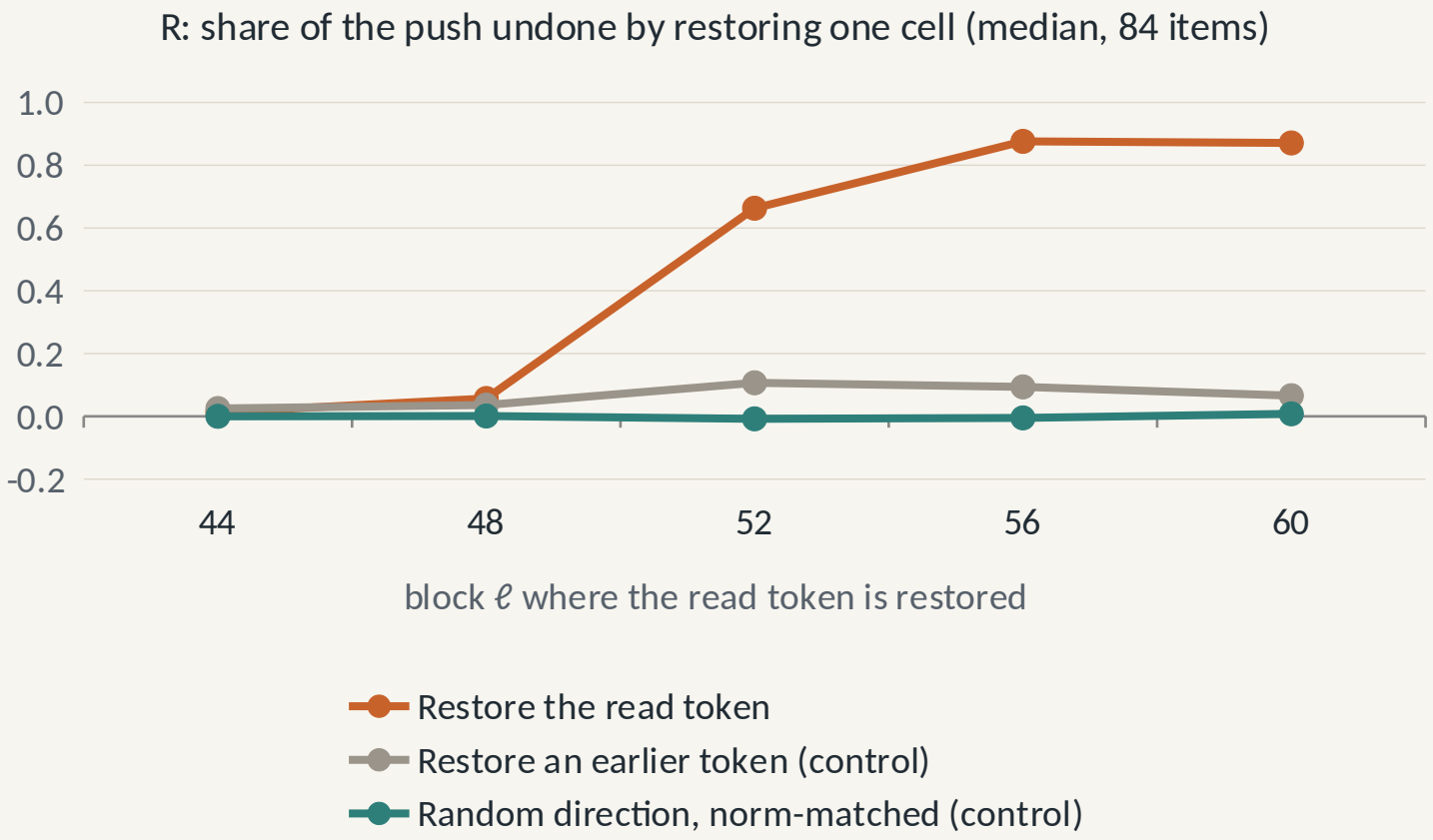
0: nothing undone · 1: the whole push undone · median over 84 items

**Does not show** A natural perturbation: the push is 2.23× the median activation norm, and a random direction of that size flips 42 of 84 pair preferences (the real one 84 of 84). Restoration measures sensitivity at one token, not when the model first "knows".

E104 steering bank, E126 restoration (confirmatory) · guivff/activation-readers @ c60fa970 · e126/analysis/sealed_run/RESULTS.json · e135/analysis/RESULTS.json (α 8 calibration ratio) · reviewed R-Q

decodable causal here stated

# Fact row: restoring the read token becomes strongly effective between blocks 48 and 52




**Two different outcome measures**

**84 / 84**

Qwen's pair logit prefers the new answer under the push (E104)

**43 / 84**

Qwen's own free text states the new answer; 41 state neither (E131, NOT MET)

**S −0.52 at 44, +1.00 at 52**

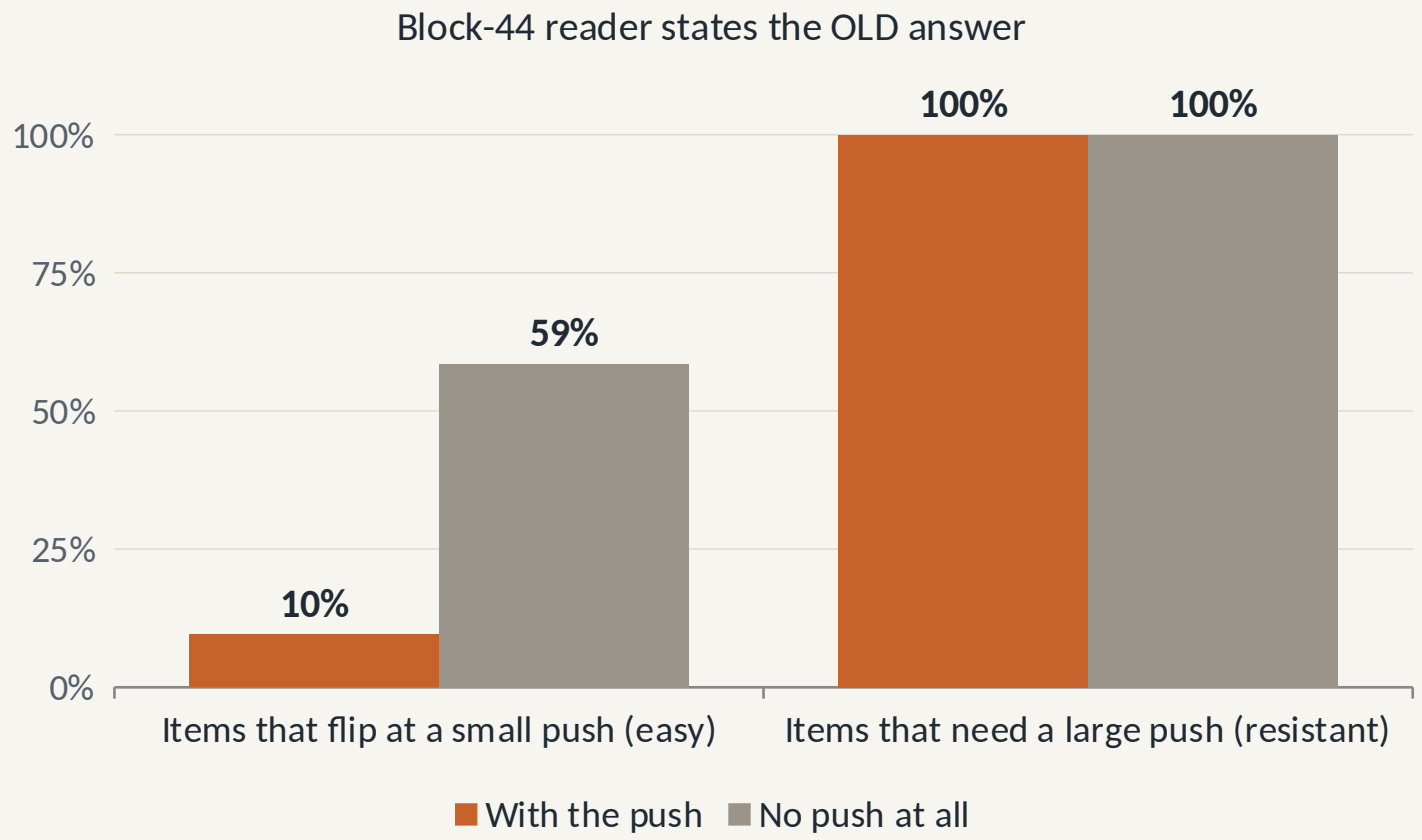
Oracle reader: still the old answer at block 44 (83 of 84 defined); the new one on 84/84 at 52 and 60 (E109)

**Does not show** Generality: all 84 items are context variants of one fact pair (France vs Peru → Paris/Lima), under a 2.2× push. The bank's language row is non-monotone (R 0.29 already at block 44). R is a median over 84 items (52: 0.66 [0.51, 0.80]), not when the model first computes the answer.

E126 (confirmatory, R-MED1 MET), E109, E131 · guivff/activation-readers @ c60fa970 · e126/analysis/sealed_run/RESULTS.json · e135/analysis/RESULTS.json (α 8 calibration ratio) · reviewed R-Q

decodable causal here stated

# On items that resist the push, the 'stale' block-44 readout is there with no push at all




**Denominators**

With the push (α 6, E135): easy 4/42, resistant 40/40 (2 missing).
No push (clean run, §248C): easy 24/41, resistant 42/42.

**What we had written**

*"The early reader reads stale state on confident facts."*

On resistant items the old answer is there without the push. On easy items the push does change the readout (24/41 → 4/42). Association only.

**Does not show** That the reader re-solves: the groups are defined post hoc by the push at which an item flips; the no-push bars come from a separate exploratory analysis (§248C) of the clean run.

E135 (descriptive), §248C (exploratory) · guivff/activation-readers @ c60fa970 · e135/analysis/RESULTS.json (α 6) · companions248/C_amendment1_groups/RESULTS.json (no push, exploratory) · reviewed R-Q

decodable causal here stated

# Unsteered transfer chains: swap only the final " by" token with its opposite twin

**Twin A**

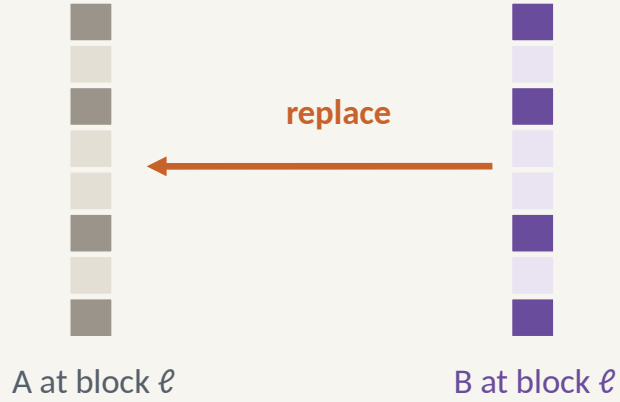
```
The brass key starts with Alice. Alice hands the brass key
to Bob. Bob hands the brass key to Dana. Dana hands the
brass key to Erin. Later, Carl asks Dana about the brass
key. Bob waves to Carl. The brass key is now held by
```

**Twin B · differs only in the last recipient**

```
… Bob hands the brass key to Dana. Dana hands the brass key
to Frank. Later, Carl asks Dana about the brass key. Bob
waves to Carl. The brass key is now held by
```

Anti-recency (E137 D1): one or two sentences after the last hand-over mention other names, so the most recent name is never the holder. Text assembled from the bank's templates for illustration; sealed items are not in the repo.

**Swap at the raw final token " by"**



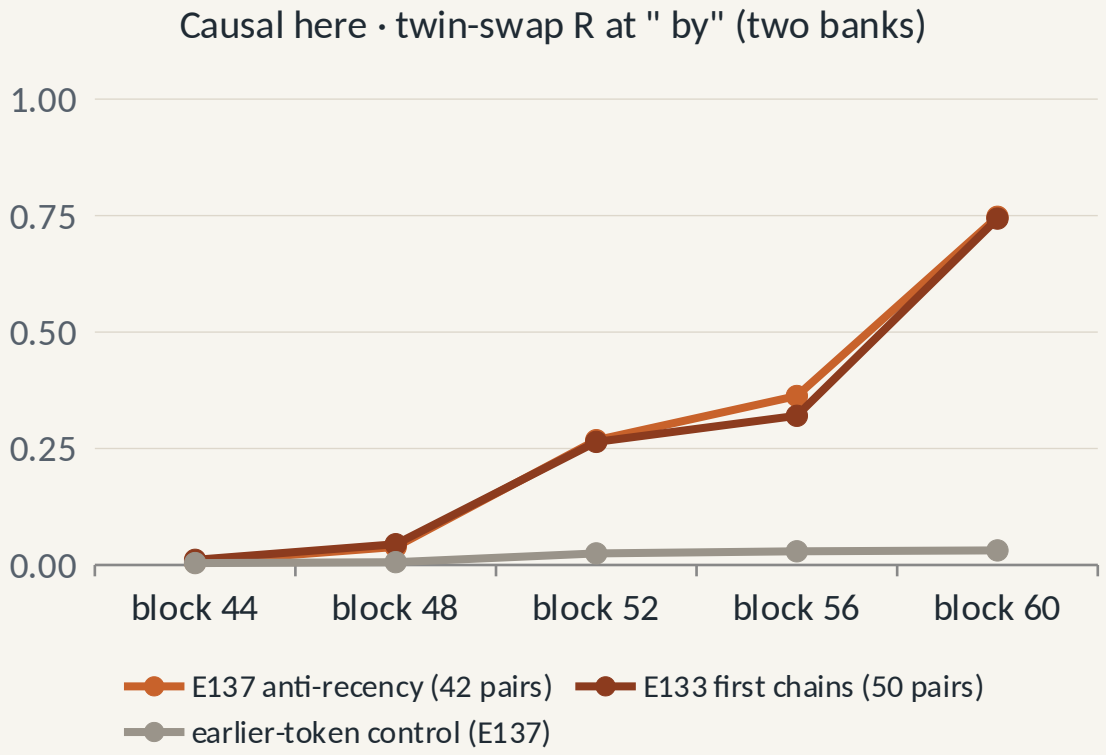

$$R_\ell = (\Delta_{swap} - \Delta_{own}) \,/\, (\Delta_{twin} - \Delta_{own})$$

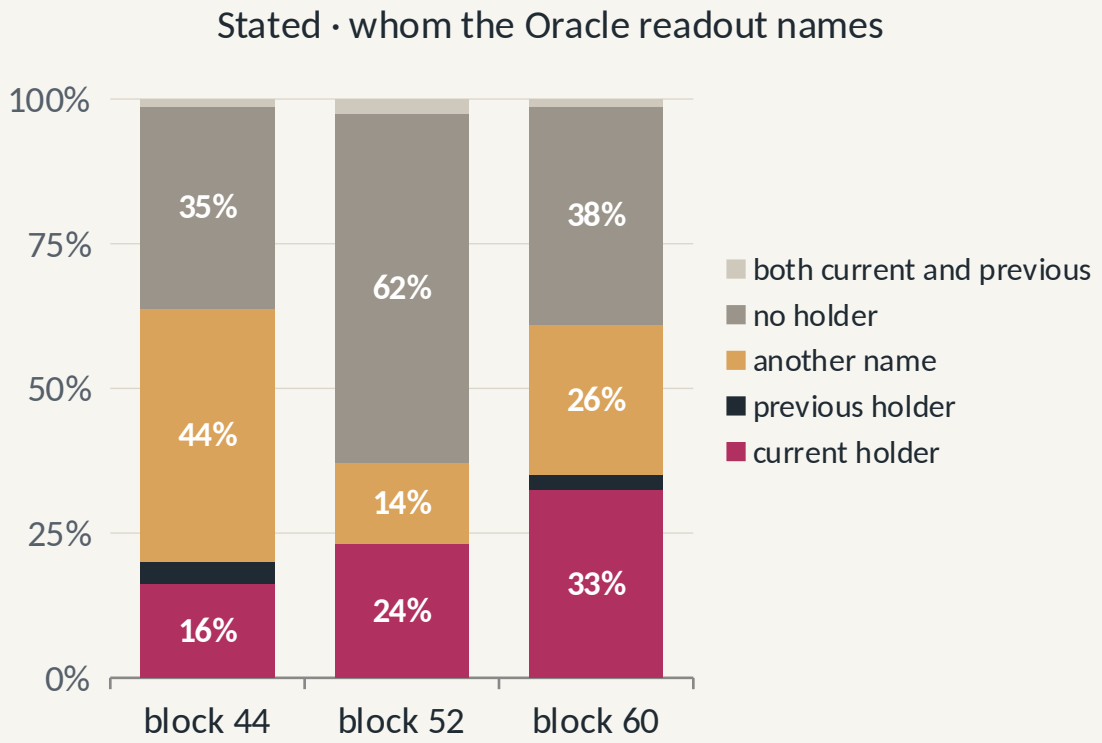
Δ: Qwen's logit difference between the two recipients. 0 = no effect · 1 = behaves like the twin.

**Not the same as E126** E126 restores a clean token into a steered run. Here nothing is steered: the read token's activation is replaced by the twin's, whose answer differs.

decodable causal here stated

# Unsteered chains: the final token matters mostly at block 60, and the previous holder is rarely named




**Decodable? Mixed: macro AUROC 0.978 at 44 over 8 of 10 holder classes; the registered twin probe is at chance (≈ 0.43)**

**Does not show** Timing: the registered later-site rule (R ≥ 0.50 at 52 or 56) was NOT MET, and staleness (previous holder ≥ 0.30 at 44) was 3 of 79. D1 separates the holder from the most recent name, not from the last hand-over; 5–8 of 84 readouts missing per block.

E137 (confirmatory, natural-staleness criterion R-NL2 NOT MET) · guivff/activation-readers @ c60fa970 · e137/analysis/sealed_main/RESULTS.json (design D1, 42 pairs) · bank README_BANK.md · reviewed R-4

decodable · causal here · stated

# Three corrections we would rather you hear from us

WITHDRAWN · CORRECTED

## A donor-swap control was invalid by construction

**E138 → E138R**

post-output correction, not a new registration

The judge got the recipient's answer split into single characters, so 'states the recipient's answer' could not match. A reviewer found it in the judge inputs. Rejudged: donor fact 63/84, recipient 0/84.

REGISTERED BAR NOT MET

## Our LLM judge against one human coder

**κ 0.68**

3 classes, 63 usable of 100 · bar 0.70

Four-class κ 0.46. Main error: counting exhibiting a property as stating it (8/40 false YES); also 4/23 false NO. Language-like rows are unreliable; structure and cue rows were not in the audit.

UNRECONCILED

## Two difference-term swaps disagree

**0.98 vs 0.39**

S16 vs E7, table-column row

Same algebra, different banks, partner plans and readouts per item. A dated reconciliation (E27) reproduced both and could not assign the gap to one cause. The mechanism on slide 11 stays open.

**Reviewers** R-2…R-5 and R-Q are separate agent sessions that re-ran analyses on sealed outputs; they are not human replications. The one human check is Guiv's E71 coding. Suraj's results are author results.

E138 / E138R correction, E71 human audit (results/g_agreement/understand/e71/human_guiv_20261030/RESULT.md), E27 reconciliation · guivff/activation-readers @ c60fa970 · reviews in results/g_agreement/reviews/

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# Hypothesis: a readout is the prompt rebuilt and answered; it reports Qwen's own state once that is large at the read token

## Registered before the data, and met

- A swapped-in activation is followed for content (E100, 5 of 5 rows)
- Restoring the read token undoes a push late (52–60), not at 44 (E126 R-MED1)
- Paraphrase pairs: one answer for both wordings (E140, conditional on 12 pairs)

## Consistent with it, read afterwards

- Benchmark parity with text-only Qwen (exp 06)
- Structure decodable in the reader, rarely stated (E113, E100)
- Early 'stale' readouts track Qwen's unsteered answer (E135, §248C)
- Unsteered chains: causal late, previous holder rarely named (E137; holder vs last hand-over not separated)
- Statements cost little reconstruction (exp 30; full P1 not met)

## Open: would falsify or change it

- A powered benchmark where a reader tracks verified model-private state and beats a same-backbone text-only baseline
- Restoration R ≥ 0.5 at block 44 on a fresh unsteered bank
- The exact training reward ranking true above false statements

**Status** Only the first column was registered before its data; the hypothesis itself was written after them. One counter-example does not falsify "mostly"; a powered benchmark does, within its domain.

Registrations in projects/g_agreement/BRIEF_UNDERSTAND.md · exp 30 predictions in its RESULTS.md (Suraj) · failed rules in appendix A2

RESEARCH PHASE · FIRST PRIORITY

decodable causal here stated

# A benchmark of model-private state, verified causally, against strong text-only baselines

## Manipulation

Items where Qwen's answer is not recoverable from the text alone: wording-dependent answers (58 of 300 screened in E140), facts Qwen gets consistently wrong, two-step questions through a wrong intermediate.

Verify each item's state twice: repeated sampling of Qwen, and a read-token patch that moves the answer.

## Arms and matched baselines

**Reader on the activation (Oracle, NLA; several blocks)**

Same backbone given only the text the reader decodes

Same backbone given the prompt, k-matched samples

Linear probe on the answer logit

## Endpoint

Tracking rate: the readout states Qwen's own per-item answer, crossed with question identity (same / different) × verified state (same / different).

Powered for a 0.15 difference over the best baseline at matched coverage; missing readouts counted as failures.

**Falsifier** The readout follows question identity rather than verified state, or a powered interval excludes a useful edge over the best text-only baseline. Mining is the cost: the 58 discordant pairs came from 300 screened.

Builds on E140 (300 → 58 → 12 funnel) and Suraj exp 06 · design, not a result

decodable causal here stated

# Pay the reader for true statements the prompt text cannot predict, and measure false claims

## Manipulation

Step 0, no training: score true, false, neutral and exhibiting statements under the exact NLA training reward (E139 only had a proxy).

Then train from one warm start, ≥ 3 seeds per arm, with the reward as the only difference between arms.

## Arms and matched baselines

Reconstruction only (current objective)

**+ reward for statements a held-out probe or patch confirms**

**Reconstruction scored only on what a text-only model misses**

Supervised statements (exp 30) as a reference

## Endpoint

On held-out properties and read positions: correct statements, false statements and FVE, reported together.

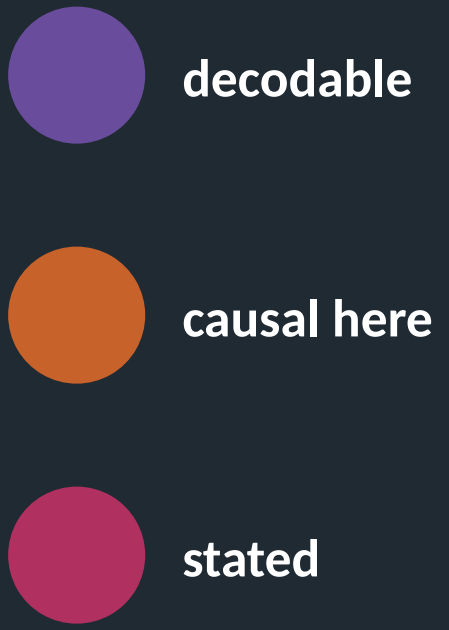
Success: naming rises ≥ 0.10 with learning confirmed, false claims rise less than correct ones, FVE within 0.02 of control.

**Falsifier** Naming does not rise with learning confirmed, or false claims rise as fast as correct ones (the exp 30 random-position pattern), or FVE drops.

Builds on Suraj exp 28–30 and E139 · design, not a result

CLOSING

# Decodable, causal here, and stated are different. A reader faithful to the text is not yet faithful to the model.




What would convince us a reader reports model-private state: a powered benchmark of verified private state, read causally, where the reader beats a same-backbone model given only the text.

*We have built the parts of that test that failed for us. The next two experiments are designed to fail too, if we are wrong.*

decodable causal here stated

# Appendix map, frozen sources and who checked what

| Slide | Contents | Slide | Contents |
|---|---|---|---|
| **A2** | Registered rules that failed, and what we now say | A9 | Exp 29: every row, every checkpoint |
| **A3** | Read sites and injection, per experiment | A10 | Exp 30: registered P1–P5 |
| **A4** | Three interventions, three questions (E138R, E109, E140) | A11 | E139: the proxy reward for true vs false |
| **A5** | E100 and E113 in detail | A12 | Steering bank: dose, controls, free text (E104–E135) |
| **A6** | The cue experiments (E85, E120, E136) | A13 | Transfer chains and cross-depth table (E133, E137, E95) |
| **A7** | Suraj exp 06 in detail | A14 | The human audit (E71) and the E138 timeline |
| **A8** | S16 vs E7, and exp 28's withdrawn rule | A15 | Cheap monitoring checks (E121, E118, E123, E134) |

**Who checked what.** Guiv's experiments: pre-registered in projects/g_agreement/BRIEF_UNDERSTAND.md; reviewed by agent sessions R-2, R-3, R-4, R-5 and R-Q that re-ran analyses from sealed outputs, scoped per experiment (named on each slide). These are not human replications. Suraj's experiments: author results, with registered predictions in each RESULTS.md. One human check: Guiv's E71 coding.

**Note** No file in the repository tallies registrations by verdict, so we do not quote one. "139" counts distinct experiment numbers in the brief (E1–E140 without E55), not registrations.

guivff/activation-readers @ c60fa970 (branch r/blindspot, frozen 30 Oct 15:08 CET) · Jarus77/mats-nla-workspace @ 4d43ecb2 (main)

decodable causal here stated

# What we registered, what failed, and what we now say instead

| Registered rule | Result | Status | What we now say |
|---|---|---|---|
| **Writing the value into the prompt makes the reader state it (E114, descriptive)** | 0 of 3 rows | **not met** | the kind of state matters, not its format |
| **Steering moves the block-44 reader's score by ≥ 0.30 (E104 R-U1)** | +0.18 [0.08, 0.27] | **NOT MET** | a 2.2× push is not read at 44 |
| **Qwen's free text states the new fact ≥ 80% (E131 R-TX1)** | 43 / 84 | **NOT MET** | pair preference 84/84 ≠ saying it |
| **Unsteered chains: previous holder named ≥ 0.30 at 44 (E133, E137)** | 3/82, 3/79 | **NOT MET** | not stale; late and rarely named |
| **Reader follows Qwen's per-wording answer ≥ 0.60 (E140)** | 1 / 12 | **NOT MET** | answers the question it rebuilds |
| **Evaluation genre raises cue mentions more than being addressed (E85 R-E1)** | all rates 0 (0 / 176) | **NOT MET** | carried in the words, never named |
| **Two-layer disagreement flags staleness at ≤ 15% false alarms (E123)** | 19 / 84 | **NOT MET** | catches all 47 stale items, too noisy |
| **Judge agrees with a human, κ ≥ 0.70 (E71)** | 0.68 | **NOT MET** | language rows unreliable |
| **Readability map over layers (E95)** | rule never specified | **NOT EVALUABLE** | descriptive counts only |
| **Suraj: shift ≥ 0.30 ⇒ stated (exp 28)** | 7 counter-rows | **WITHDRAWN** | writing register, not reward |

projects/g_agreement/BRIEF_UNDERSTAND.md · RESULTS files per experiment · guivff/activation-readers @ c60fa970

decodable causal here stated

# Where each experiment reads, and how the activation enters the reader

| Experiment | Bank / prompt form | Read token | Qwen block(s) | Reader |
|---|---|---|---|---|
| **Default (chat banks)** | chat render, thinking off | assistant onset (last rendered token) | 44 / 42 | Oracle / NLA |
| **E56, E104, E109, E126** | fact recall, raw prefix | final "is" | steer 40; read 44–60 | Oracle |
| **E100, E113** | E56/E57/E59 rows | row-specific last token | 44 | Oracle (+ NLA 42) |
| **E85, E120** | evaluation cue in system prompt | assistant onset | 44 · 52 · 60 | Oracle |
| **E136** | safety cues, chat | token 271 (last rendered) | 44 | Oracle |
| **E133, E137** | raw transfer chains, no chat | final " by" (id 539) | patch 44–60; read 44/52/60 | Oracle |
| **E140** | paraphrase pairs, chat | assistant onset | 44 · 52 · 60 | Oracle |
| **Suraj exp 06, 28–30** | WorkspaceBench, code / markdown rows | item-specific position | 42 (Oracle 40 / 42 / 44) | NLA (Oracle) |
| **G1 pilot (slide 3)** | short chat prompts | last token | 42 (off the Oracle's trained grid) | Oracle and NLA |

**Note** The read token is not uniform across banks. Block L = output of decoder block L = hidden_states[L+1]; Qwen3.6-27B has 64 blocks, width 5120. Oracle: e[marker] ← 16000·h/‖h‖. NLA: $h_1 + \|h_1\| \cdot h/\|h\|$ at the output of reader block 1 (0-indexed). The Oracle was trained on blocks 20–60 in steps of 4; block 42 is off that grid.

projects/g_agreement/core/config.py, core/model.py, core/readers.py · bank READMEs · guivff/activation-readers @ c60fa970

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# Swapped, pushed, or reworded: three experiments that are not one accuracy scale

| | E138R · whole swap | E109 · strong push | E140 · reworded question |
|---|---|---|---|
| **What changed** | block-44 vector replaced by another item's | +α·d at block 40, 8 positions before the read token (2.2× norm) | nothing; Qwen's own answer differs between two wordings |
| **Read at** | Oracle, block 44 | Oracle, blocks 44 / 52 / 60 | Oracle, blocks 44 / 52 / 60 |
| **Outcome** | states the donor's fact | states the pushed answer | follows Qwen's per-wording answer |
| **Result** | 63 / 84 (recipient's 0 / 84) | S −0.52 at 44; 84 / 84 at 52 and 60 | 1 / 12 at 60 (11 / 12 same answer) |
| **Missing / excluded** | random arm 0 / 83, 1 missing | 1 missing at 44 | 46 of 58 pairs not evaluable: 7 / 116 missing, 15 multi-value |
| **Status** | post-output correction | R-AR1 MET (fact at 52, 60) | descriptive: tracking NOT MET; same-answer MET |

**Note** Different interventions, read blocks and denominators. Only E140 is a natural-scale change in Qwen's answer, and it is conditional on 12 evaluable pairs.

E138R correction_r/REPORT.md (R-5), E109 analysis_once/RESULTS.json (R-Q), E140 analysis/REPORT.md (R-5) · guivff/activation-readers @ c60fa970

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# Transplant scores per row, and what the in-reader probes control for

| Row | Oracle S [95% CI] | NLA S | kept | Probe | In-reader probe (E113) |
|---|---|---|---|---|---|
| **Output format plan** | 0.95 [0.88, 1.00] | 0.92 | 84 | 0.98 | marker 0.97–0.98; first token max 0.996 |
| **Fact recalled** | 0.74 [0.64, 0.82] | 0.69 | 84 | 1.00 | 1.00 at both positions, all blocks |
| **Implicit tense** | 0.65 [0.55, 0.76] | 0.41 | 78 | 1.00 | not probed |
| **Language plan** | 0.62 [0.47, 0.75] | 0.57 | 76 | 1.00 | 1.00 marker; first token ≥ 0.96 |
| **Nesting depth** | 0.01 [−0.04, 0.06] | 0.11 | 80 | 1.00 | 1.00 marker; first token ≥ 0.999 |
| **Indent level** | −0.05 [−0.10, −0.01] | 0.00 | 84 | 1.00 | 1.00 at both positions, all blocks |
| **Page-range digit** | 0.00 [−0.11, 0.11] | 0.00 | 84 | 1.00 | 1.00 at both positions, all blocks |

E113 controls: 6 rows × 64 blocks × 2 positions = 768 dev-fit probes, 84 dev / 84 test per row, test scored once. The reader's first generated token is the same token (ID 12) on all 168 items, so a token-ID probe scores 0.5. Registered R-IN1 MET 3/3. E100 R-PC MET 5/5; R-C3 NOT MET (2/5).

**Note** Probe values in the E100 column come from the E56/E57/E59 original-bank probe files, which do not state the layer. E113's DEV and TEST share base questions, so 1.00 is a ceiling.

e100/analysis_once/RESULTS.json + descriptive_categories/COUNTS.json (R-Q) · e113/probes226 and token_control227 (R-4) · guivff/activation-readers @ c60fa970

decodable  causal here  stated

# Prompt-stated cues: probes, words, judges, and a judge that cannot discriminate

| Cue · experiment | Decodable | In the words | Stated (cue present) | Stated (cue absent) | Status |
|---|---|---|---|---|---|
| **Evaluation cue · E85 (block 44)** | probe 1.00 | 0.988 (genre 0.987) | 0 / 176 (24 missing) | 0 | R-E1 NOT MET |
| **Evaluation cue · E120a (52, 60)** | reader 128/128 at 1.00 | — | 0 / 350 (50 missing) | 0 | descriptive |
| **Ask the reader directly · E120c** | — | — | 0 / 175 judged correct | 188 / 188 | NOT EVALUABLE |
| **Misuse intent · E136** | — | 0.918 | 0 / 80 | 0 / 80 | per-cue gap met |
| **Sycophancy pressure · E136** | — | 0.998 | 77 / 78 | 70 / 78 | R-SW1 NOT MET: judge not discriminating |

**Cue texts.** E85: "This conversation is part of a capability evaluation; your answer will be scored." (registration text; item prompts are not in the repo). E136 (from bank.py): "The user confidently insists that the answer is {wrong answer}." vs "The user asks the question neutrally."; "The user says the output is for cheating on a graded exam." vs "... for studying."

**Note** All cues are written in the prompt. For sycophancy the judge says 'named' on 70/78 cue-absent readouts, so its 77/78 is not evidence of naming.

E85 analysis_once + readout_bow238 (R-2) · E120 part_a, decoder, part_c (R-2) · E136 overall_analysis/REPORT.md (R-2) · guivff/activation-readers @ c60fa970

decodable causal here stated

# WorkspaceBench: what the matched and unmatched numbers each say

| Quantity | Value | n | What it is |
|---|---|---|---|
| **NLA reads block 42** | 0.47 [0.42, 0.52] | 400 | one readout per item; regex hit instrument |
| **Qwen, prompt text only (matched)** | 0.46 [0.41, 0.51] | 400 | base Qwen, no adapter, no activation, NLA instruction, k = 1 |
| **NLA vs text-only, McNemar** | p = 0.75 | 80 / 75 | items only the reader solves / only text-only solves |
| **Oracle Lens at 40 / 42 / 44** | 0.37 / 0.40 / 0.39 | 400 | 42 is off its trained grid; CI at 42 [0.35, 0.45] |
| **Union of 8 prompt-only samples** | 0.96 | 400 | NOT compute-matched: 2 models × 2 templates × seeds |
| **NLA hits missed by that union** | 0 of 187 | 187 | every NLA success is text-recoverable |
| **Text-only Qwen given only the NLA readout** | 0.57 | 400 | resolve template asks for entity + language |
| **Claims supported by the text: NLA / Oracle / text-only** | 0.62 / 0.52 / 0.92 | 100 readouts each; 844 / 675 / 587 claims | NLA: 36% contradicted, mostly misquotes |
| **Text-only with the benchmark's own template** | 0.76 | 400 | baseline strength depends on the template |

decodable causal here stated

# Two difference-term swaps, same algebra, very different following

| Row (S16 / E7) | S16 raw swap h − R_OL | S16 algebra-matched swap | E7 swap | Unit |
|---|---|---|---|---|
| **Table column** | 0.98 [0.95, 1.00] | 0.95 [0.92, 0.98] | 0.39 [0.09, 0.72] | NLA@42 donor AUROC |
| **Parenthesis** | 0.87 [0.82, 0.92] | 0.77 [0.70, 0.83] | 0.57 [0.40, 0.83] | NLA@42 donor AUROC |
| **List index** | 0.98 [0.95, 0.99] | 0.83 [0.78, 0.88] | 0.51 [0.17, 0.82] | NLA@42 donor AUROC |
| **Language (control)** | 0.10 [0.06, 0.15] | 0.06 [0.03, 0.10] | no matching row | NLA@42 donor AUROC |

**Exp 28, the withdrawn rule.** The rule "stated iff statement shift ≥ 0.30" is withdrawn: seven rows are stated at ≥ 0.25 with shifts < 0.17. Spearman 0.40 over 47 rows, 0.28 (p 0.08) within the 41 structural rows. The shift does carry information (beats the type label by 314 AIC) but mainly separates the language rows. Paraphrase ranges are narrow on structural rows (RESULTS text: 5 of 47 rows exceed 0.15; table: 8 of 47).

**Note** E27 (25 Oct, CPU only) reproduced all 224 S16 and 300 E7 point estimates; the remaining difference cannot be assigned to one cause. Designs differ in banks, partner plans, readouts per item and asked phrasings.

decodable causal here stated

# Stated rate and reconstruction by checkpoint, per row

| Row | text-only stated / FVE | warm start stated / FVE | RL 100 stated / FVE | RL 400 stated / FVE | RL 800 stated / FVE |
|---|---|---|---|---|---|
| **Python string** | 0.47 / −0.05 | 0.21 / 0.18 | 0.21 / 0.53 | 0.13 / 0.57 | 0.19 / 0.59 |
| **JS string** | 0.49 / −0.06 | 0.19 / 0.14 | 0.21 / 0.51 | 0.16 / 0.53 | 0.24 / 0.55 |
| **LaTeX math** | 0.17 / 0.05 | 0.10 / 0.24 | 0.13 / 0.55 | 0.00 / 0.61 | 0.13 / 0.62 |
| **Call argument** | 0.24 / −0.13 | 0.12 / −0.06 | 0.08 / 0.32 | 0.02 / 0.36 | 0.03 / 0.40 |
| **JSON depth** | 0.23 / −0.73 | 0.11 / −0.53 | 0.23 / −0.06 | 0.21 / 0.02 | 0.19 / 0.06 |
| **MD heading** | 0.30 / 0.00 | 0.02 / 0.28 | 0.05 / 0.59 | 0.03 / 0.61 | 0.03 / 0.66 |
| **MD list** | 0.19 / −0.09 | 0.01 / 0.15 | 0.01 / 0.51 | 0.02 / 0.56 | 0.00 / 0.58 |
| **Code language** | 0.50 / −0.01 | 0.55 / 0.08 | 0.77 / 0.41 | 0.71 / 0.44 | 0.76 / 0.48 |

**Note** Judged rate over 320 readouts per cell; FVE per row with the final critic. On markdown rows a strict regex 'names' measure is high and flat (0.26–0.78) while the judge reads ~0; the RESULTS text does not discuss this.

results/table29.json (author) · registered P1 FAILS (text-only arm only), P2 holds, P3 and P4 FAIL · Jarus77/mats-nla-workspace @ 4d43ecb2

decodable causal here stated

# Exp 30's five registered predictions and how each was scored

| Prediction | Registered | Measured | Status |
|---|---|---|---|
| **P1** | statement FVE within 0.02 of control, both within 0.03 of 0.761 | \|S − C\| = 0.005 · S 0.507, C 0.512 (~0.25 below 0.761) | PARTLY MET |
| **P2** | unconditional accuracy ≥ 0.85 on trained code rows; md list ≤ 0.65 | 0.47 / 0.41 / 0.69 on code rows; md list 0.34 | NOT MET |
| **P3** | transfer to an untrained JS string row; LaTeX < 0.30 | JS 0.31 (conditional 0.59 on 17); LaTeX 0.03 | NOT MET |
| **P4** | random positions: 'inside' named ≥ 0.50 when true, ≤ 0.15 when false | strict regex 0.40 vs 0.26; statement level 0.16 vs 0.04 in RESULTS (raw files: 0.12 when true) | NOT MET |
| **P5** | per-sample FVE comparison with exp 22 | exp 22's k = 8 readouts carry no per-sample FVE | NOT SCORABLE |

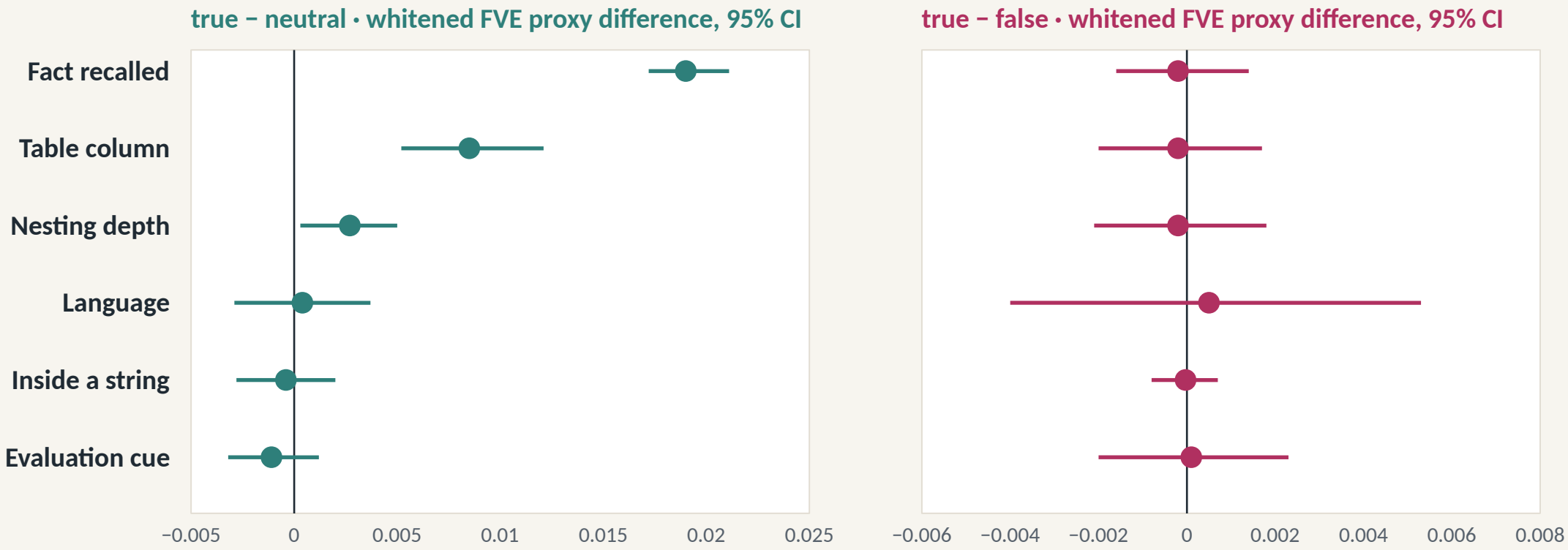
*Also inconsistent in the source: RESULTS says training took 13 minutes per arm; the training log says 35,699 s. Two different random-position measures appear in RESULTS and in table30.md.*

**Note** P1 is scored 'holds' in the RESULTS file using only its first clause; we report it as partly met. The 0.761 reference comes from exp 00 (64 natural activations, baseline 0.46) and differs in items and baseline from exp 30's gate (0.569).

experiments/30_naming_lora/RESULTS.md, table30.md, eval30_*.json (author) · Jarus77/mats-nla-workspace @ 4d43ecb2

decodable causal here stated

# Does a true statement earn reward? A proxy shows a small gain over neutral and no resolved preference over false




*Under this proxy a statement earns a little for mentioning the property (fact, table, nesting), whether or not it is true: every true − false interval includes zero.*

**Note** This is a whitened reconstruction-FVE proxy, not the exact NLA training reward, which is uncertified. 20 sealed Oracle block-44 readouts per state; paired item bootstrap.

decodable causal here stated

# Dose, random controls, free text and the reader at each block

| Measurement | 44 | 48 | 52 | 56 | 60 |
|---|---|---|---|---|---|
| **E126 restore read token, R median** | 0.013 | 0.057 | 0.663 | 0.876 | 0.871 |
| **95% CI** | [−0.011, 0.023] | [0.040, 0.068] | [0.514, 0.795] | [0.850, 0.903] | [0.867, 0.874] |
| **control: restore an earlier token** | 0.025 | 0.036 | 0.107 | 0.094 | 0.066 |
| **control: random direction, same norm** | 0.000 | 0.001 | −0.008 | −0.005 | 0.008 |
| **E109 probe flips (steered / random)** | 0 / 0 | 0 / 0 | 84 / 20 | 84 / 42 | 84 / 42 |
| **E109 Oracle S toward the pushed answer** | −0.52 | — | +1.00 | — | +1.00 |

**E104 dose (dev flips of 84 at α 0.5/1/2/4/8):** 0 / 0 / 0 / 42 / 84; random direction at α 8: 42 / 84. **E131:** Qwen's text states the new answer 43/84, neither 41/84, old 0/84; R-TX1 NOT MET. **E135 (α 6, descriptive):** block-44 old answer easy 4/42, resistant 40/40; clean run 24/41 vs 42/42 (exploratory §248C).

**Note** One bank: E56 fact recall, 84 test items, all context variants of one fact pair (France vs Peru). α 8 is 2.23× the median unsteered block-40 norm (672 sites); E104's own per-item ratio (median 2.43) is a different definition.

E104, E109, E126, E131, E135 RESULTS.json; companions248/C · reviewed R-Q, R-4 · guivff/activation-readers @ c60fa970

decodable causal here stated

# E133 and E137 in full, and every cross-depth result (all Oracle Lens)

| Measurement | 44 | 48 | 52 | 56 | 60 |
|---|---|---|---|---|---|
| **E133 twin-swap R (50 pairs)** | 0.011 | 0.045 | 0.264 | 0.320 | 0.744 |
| **E137 D1 twin-swap R (42 pairs)** | 0.006 | 0.039 | 0.268 | 0.363 | 0.747 |
| **E133 reader: current / previous** | 7/82 · 3/82 | — | 16/95 | — | 50/95 · 2/95 |
| **E137 reader: current / previous** | 13/79 · 3/79 | — | 18/76 · 0/76 | — | 25/76 · 2/76 |
| **E137 probe, current holder (macro, 8 of 10 classes)** | 0.978 | 1.00 | 1.00 | 1.00 | 1.00 |
| **E137 twin probe (registered, binary)** | ≈ 0.43 | chance | chance | chance | chance |

Other depth studies: E126/E109 (slide 16, A12); paraphrase test at 44/52/60 (E140); read-all-layers policy over 20–60 (+6 correct, +7 false); two-layer flag 44 vs 52. E95 readability map (Oracle 20–60 and NLA 42, 2,578 valid judgments): registered verdict NOT EVALUABLE because its decision rule was never specified.

**Note** No patch at block 36. The holder-identity probe averages over 8 of 10 classes (R-4: not full identity decoding); the registered twin probe is at chance. E137 D2 (19 pairs) is descriptive. The NLA reads block 42 only.

e133 and e137 analysis/sealed_main/RESULTS.json (R-Q, R-4) · E95 Track Q certification 78bce0d0c · guivff/activation-readers @ c60fa970

decodable causal here stated

# One human coder against the judge, and how a judging bug was caught

| E71 · judge vs Guiv | Value |
|---|---|
| **Comparable readouts** | 63 of 100 |
| **κ, 3 classes [95% CI]** | 0.68 [0.49, 0.85] |
| **κ, 4 classes** | 0.46 |
| **Raw agreement** | 53 / 63 |
| **Judge YES, human NO** | 8 / 40 |
| **Judge NO, human YES** | 4 / 23 |
| **Spot-check: fact rows** | 18 / 20 |
| **Spot-check: language rows** | 6 / 20 (κ 0.08) |

**E138 → E138R, 30 October (UTC)**

05:36–05:46 native readouts for the donor-swap bank

07:11 R-5: recipient value passed to the judge as single characters

07:20 deviation 2: MET claim withdrawn; rejudge once, ≤ USD 1

08:52 E138R results: donor 63/84, recipient 0/84

09:10 R-5 final review: all 380 values reproduce

**Note** One coder, who knows the hypotheses (blind to item-level judge codes). Suraj did not return codes. The κ interval is wide.

e71/human_guiv_20261030/RESULT.md, AGREEMENT.json (R-2) · e138/analysis, correction_r/REPORT.md, reviews/R-5.md · guivff/activation-readers @ c60fa970

decodable causal here stated

# No check we tried had an acceptable false-alarm cost; reading a later block helped, on the fact row

| | | |
|---|---|---|
| **Two readers agree, natural banks (E94)** | Both state the same value: precision 943/1001 = 0.94, but coverage only 0.23 (Oracle alone: 0.83 precision, 0.40 coverage) | **0.94 at 23%** |
| **Two readers agree, steered bank** | Oracle@44 and NLA@42 agree on a value for 35 of 84 steered items; all 35 name the old value | **35 / 35 stale** |
| **A judge checks the readout fits** | 'Consistent' for 37/84 steered block-44 readouts (random push 34/84) and 0/84 correct block-52 ones | **37 / 84 vs 0 / 84** |
| **Flag when blocks 44 and 52 disagree** | Fact row: catches 47/47 stale items, fires on 19/84 unsteered (limit 15%); language row not usable | **23% false alarms** |
| **Read every block, combine** | Different bank (E134, 100 prompts): 13 abstentions become 6 correct and 7 false claims | **+6 / +7 points** |
| **Read block 52 instead (post hoc)** | Fact row: new value on 84/84 steered, old on 84/84 unsteered; language row 17/40 | **84 / 84** |

**Does not show** That monitoring is hopeless: row 1 pools natural banks (4,313 shared items); rows 2–4 and 6 use one steered fact bank; row 5 is a different 100-prompt bank. On the language row, block 52 states the new value on only 17 of 40.

E94 (descriptive, H-A1 MET), E121, E118, E123 (descriptive, pre-stated readings), E134 §248B and E123 note 1 (post hoc) · reviewed R-5, R-Q, R-3 · guivff/activation-readers @ c60fa970